

Complex Analysis

5.2 Euler's equation

$$e^{i\theta} = \cos \theta + i \sin \theta, e^{-i\theta} = \cos \theta - i \sin \theta$$

$$Z = re^{i\theta} = r[\cos \theta + i \sin \theta] \Rightarrow \text{Polar form}$$

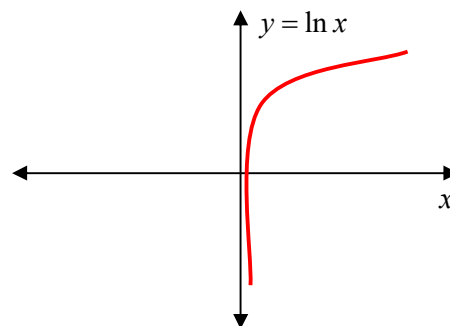
$$e^{\theta} = 1 + \theta + \frac{\theta^2}{2!} + \frac{\theta^3}{3!} + \dots, e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \dots = 1 + i\theta - \frac{\theta^2}{2!} - \frac{\theta^3 i}{3!} + \frac{\theta^4}{4!} \dots$$

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \dots\right)$$

$$(\cos x + i \sin x)^n = (e^{ix})^n \Rightarrow e^{inx} = \cos nx + i \sin nx$$

$$e^{i\pi} = \cos \pi + i \sin \pi \Rightarrow e^{i\pi} = -1$$

$$\log e^{i\pi} = \log(-1) \Rightarrow i\pi = \log(-1)$$



$$Z = x + iy, \quad xy \in \mathbb{R} \quad \left[\begin{array}{l} \text{Real } z = x \\ \text{Imaginary } z = y \end{array} \right]$$

$$Z = r(\cos \theta + i \sin \theta)$$

$$\operatorname{Re} Z = r \cos \theta, \quad \operatorname{Im} Z = r \sin \theta$$

$$\int e^{i\theta} d\theta = \frac{e^{i\theta}}{i} = \frac{\cos \theta + i \sin \theta}{i} = -i \cos \theta + \sin \theta$$

$$\Rightarrow \int \cos \theta d\theta = \operatorname{Re} \int e^{i\theta} d\theta = \sin \theta, \quad \int \sin \theta d\theta = \operatorname{Im} \int e^{i\theta} d\theta = -\cos \theta$$

Example : $E = E_0 \sin \omega t = E_0 e^{i\omega t}$

$$\operatorname{Re}(e^{i\omega t}) = \cos \omega t \quad \& \quad \operatorname{Im}(e^{i\omega t}) = \sin \omega t$$

So we will take an imaginary point.

$$z = x + iy \Rightarrow \theta = \text{Argument} = \tan^{-1} \left(\frac{y}{x} \right)$$

$$|z| = \text{modulus} = r = \sqrt{x^2 + y^2}, \quad \theta = \arg z \pm 2n\pi$$

$$-\pi < \arg z < \pi \Rightarrow \text{Principle Argument}$$

$$\sin n\pi = 0, \quad n = 0, 1, 2, 3, \dots$$

$$\cos n\pi = (-1)^n, \quad n = 0, 1, 2, 3, \dots$$

Cyclic functions:

$$\sin ix = \frac{1}{2i} [e^{i(ix)} - e^{-i(ix)}] = \frac{i}{2} [e^x - e^{-x}] = i \sinh x$$

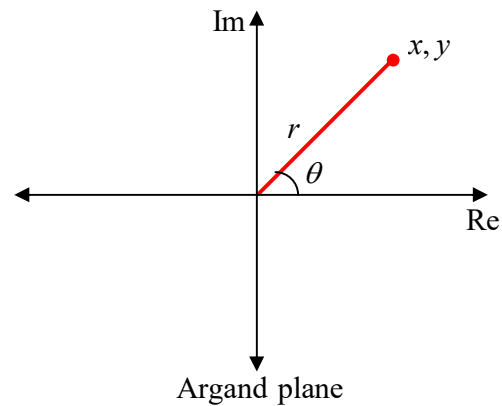
$$\cos ix = \frac{1}{2} [e^{i(ix)} + e^{-i(ix)}] = \frac{1}{2} [e^x + e^{-x}] = \cosh x$$

$$e^{ix} = \cos x + i \sin x, \quad e^{-ix} = \cos x - i \sin x$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}, \quad \sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\int e^{ax} \sin bxdx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx] \quad (\text{Imaginary part})$$

$$\int e^{ax} \cos bxdx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx] \quad (\text{Real Part})$$



Roots of the equation: Four Numbers

$$1 = e^{i2n\pi}, \quad -1 = e^{i[2n+1]\pi}, \quad i = e^{i\left[2n+\frac{1}{2}\right]\pi}, \quad -i = e^{i\left[2n+\frac{3}{2}\right]\pi}$$